Labeling Chaos to Learning Harmony: Federated Learning with Noisy Labels

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01 Context & approach

The challenge

Devices can hold labels of very different quality. *FedLN* offers three alternative ways to handle this noise: correct labels before training, add teacher guidance during local learning, or reduce noisy clients' influence during aggregation.

Our contribution

Estimate client label noise from embeddings or prediction confidence.
Offer three alternatives: *NNC*, *AKD* and *NA-FedAvg*.
Test diverse synthetic and human-annotated noise in vision and audio.

A NNC: correct labels

Use a pretrained encoder and local nearest-neighbor voting to replace mismatched labels before federated training.

B AKD: guide local learning

Use estimated noise to activate extra teacher supervision from pretrained embeddings or global-model outputs.

C NA-FedAvg: reweight clients

Estimate noise once after warm-up, then give noisier clients less weight during server aggregation.

Scope

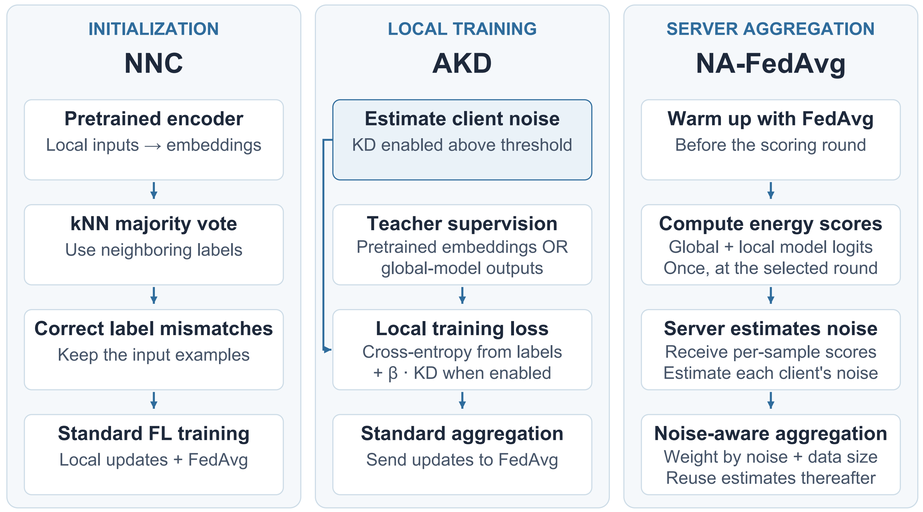
The best intervention depends on the noise pattern; *NNC* can fail under class flipping. Pretrained encoders add cost; three-run tables omit dispersion and do not establish equal computation.

02 Method

Three alternative interventions for client-dependent label noise in federated learning.

Three alternatives for handling noisy labels

Each acts at a different stage of federated learning.



KD = knowledge distillation

Core formulation

$$\mathcal{L}_{\text{AKD}} = \mathcal{L}_{\text{CE}} + \beta u_{\varepsilon}(n_l^m)\mathcal{L}_{\text{KD}}$$

For client m, estimated noise n_l^m gates teacher guidance through u_ε. β controls its strength alongside label-based cross-entropy.

$$\gamma_m = (1 - n_l^m)\frac{N_m}{N}$$

NA-FedAvg multiplies client m's data share N_m/N by its estimated clean-label fraction $1\text{-}n_l^m$.

Paper Code

03 Results

74.76%
NNC under random noise
CIFAR-10; 70% noise

54.59%
NNC under class flipping
CIFAR-10; 70% noise

Accuracy depends on the noise pattern

Method	Random noise (%)	Class flipping (%)
FedAvg	57.65	63.54
FedCorr	67.27	65.23
NNC (Ours)	**74.76**	**54.59**
AKD · embeddings (Ours)	**68.43**	**68.39**
NA-FedAvg (Ours)	**65.34**	**67.92**

CIFAR-10/ResNet-20: 70% label noise on 80% of 30 clients; random versus class-flipping noise. Three-run mean accuracy after 200 rounds.

Dependence on clients with clean labels

Noisy clients	NNC (%) (Ours)	NA-FedAvg (%) (Ours)
25%	**73.80**	**75.99**
50%	**74.54**	**73.27**
75%	**73.50**	**70.99**
100%	**73.76**	**64.02**

CIFAR-10: 40% random label noise within each noisy client; 30 clients, 200 rounds, three-run mean accuracy.

Takeaway

No single intervention wins everywhere: *NNC* leads the selected random-noise comparison but falls below FedAvg under class flipping. *NA-FedAvg* becomes less effective as **clean clients disappear**.